Ecuaciones principales

$$F_a(t) = F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$F_Z(t) = \frac{P_a(t) - P_p(t)}{Z}$$

$$F_L(t) = \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt$$

$$F_C = C\,\frac{dP_p(t)}{dt}$$

$$F_R(t) = \frac{P_p(t)}{R}$$

Transformada de Laplace

$$F_a(s) = F_Z(s) + F_L(s) = F_C(s) + F_R(s)$$

$$F_Z(s) = \frac{P_a(s) - P_p(s)}{Z}$$

$$F_L(s) = \frac{P_a(s) - P_p(s)}{Ls}$$

$$F_C(s) = C_s P_p(s)$$

$$F_R(s) = \frac{P_p(s)}{R}$$

$$\frac{P_a(s) - P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{Ls} = C_s P_p(s) + \frac{P_p(s)}{R}$$

Procedimiento algebraico.

$$\frac{P_a(s)}{Z} - \frac{P_p(s)}{Z} + \frac{P_a(s)}{Ls} - \frac{P_p(s)}{Ls} = C_s P_p(s) + \frac{P_p(s)}{R}$$

$$\left(\frac{1}{Z} + \frac{1}{Ls}\right) P_a(s) = \left(C_s + \frac{1}{R} + \frac{1}{Z} + \frac{1}{Ls}\right) P_p(s)$$

$$\frac{Ls + Z}{LZs} P_a(s) = \frac{CLRZs^2 + L(Z+R)s + RZ}{LRZs} P_p(s)$$

$$\frac{P_P(s)}{P_a(s)} = \frac{\frac{Ls+Z}{LZs}}{\frac{CLRZs^2 + L(Z+R)s + RZ}{LRZs}}$$

Resultado de la función de transferencia

$$\frac{P_P(s)}{P_a(s)} = \frac{LRs + RZ}{CLRs^2 + L(Z+R)s + RZ}$$

Error en estado estacionario

$$e(s) = \lim_{s\to 0} sP_a(s)\left(1 - \frac{P_P(s)}{P_a(s)}\right) = \lim_{s\to 0} s\,\frac{1}{s}\left(\frac{LRs + RZ}{CLRs^2 + L(Z+R)s + RZ}\right) = 0$$

$$e(t) = 0V$$

Estabilidad del sistema en lazo abierto

Para analizar la estabilidad del sistema en lazo abierto.

$$CLRZs^2 + L(Z+R)s + RZ = 0$$

a b c

$$\lambda_{1,1} = \frac{-L(Z+R) \pm \sqrt{L^2(Z+R)^2 - 4(CLR^2Z^2}}{2\,CLRZ}$$

Hipotenso

$Z = 0.010$

$L = 0.001$

$R = 1.000$

$C = 3.500$

$$\frac{P_P(s)}{P_a(s)} = \frac{LRs + RZ}{CLRZs^2 + L(R+Z)s + RZ} \simeq \frac{0.001s + 0.01}{3.5\times10^{-5}s^2 + 0.00101s + 0.01}$$

$\lambda_1 = -14.429 + 8.8051i,$

$\lambda_2 = -14.429 + 8.8051i,$

Normotenso

$Z = 0.010$
$L = 0.001$
$R = 1.000$
$C = 1.500$

$$\frac{Pp(s)}{Pa(s)} = \frac{0.001 + 0.01}{1.5\times10^{-5}s^2 + 0.00101 + 0.01}$$

$\lambda_1 = -12.062,$
$\lambda_2 = -55.272,$

Estable y sobreamortiguada

Hipertenso

$Z = 0.010$
$L = 0.001$
$R = 1.000$
$C = 1.000$

$$\frac{Pp(s)}{Pa(s)} = \frac{0.001 + 0.01}{0.000013^2 + 0.00101s + 0.01}$$

$\lambda_1 = -10$
$\lambda_2 = -10$

Estable y críticamente amortiguada.

Modelo de ecuaciones integro-diferenciales.

$$F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$\frac{Pa(t) - Pp(t)}{Z} + \frac{1}{L}\int[Pa(t) - Pp(t)]dt = C\frac{dPp(t)}{dt} + \frac{Pp(t)}{R}$$

La única variable desconocida es la salida $Pp(t)$.

$$\frac{Pa(t)}{Z} + \frac{1}{L}\int[Pa(t) - Pp(t)]dt - C\frac{dPp(t)}{dt} = \frac{Pp(t)}{R} + \frac{Pp(t)}{Z}$$

$$Pp(t)\left(\frac{1}{R} + \frac{1}{Z}\right) = \left[\frac{Pa(t)}{Z} + \frac{1}{L}\int[Pa(t) - Pp(t)]dt - C\frac{dPp(t)}{dt}\right]$$

$$Pp(t)\frac{Z+R}{RZ} = \left[\frac{Pa(t)}{Z} + \frac{1}{L}\int[Pa(t) - Pp(t)]dt - C\frac{dPp(t)}{dt}\right]$$

$$Pp(t) = \left[\frac{Pa(t)}{Z} + \frac{1}{L}\int[Pa(t) - Pp(t)]dt - C\frac{dPp(t)}{dt}\right]\frac{RZ}{Z+R}$$